

# Network monitoring

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## 1 Introduction

2 Here, we used simulations to test the efficiency of monitoring species interactions and species  
3 richness within BONs, testing several methods for design and optimization.

- 4 • Context

- 5 ▶ Biodiversity monitoring should encompass all facets of biodiversity
- 6 ▶ Species interactions are rarely considered in comparison to species richness metrics,  
7 in part due to their inherent sampling difficulty
- 8 ▶ Yet, interactions represent an biodiversity component which should be considered  
9 when implementing monitoring networks
- 10 ▶ As interactions are inherently challenging to sample, developing methods for opti-  
11 mizing strategies is even more relevant.

- 12 • Key questions

- 13 ▶ How should we implement monitoring programs *across space* to efficiently capture  
14 network composition and changes as essential components of biodiversity?
- 15 ▶ Can we monitor species interactions as efficiently as species ranges? What would  
16 be realistic expectations for monitoring species interactions?

## 17 Methods

18 We explored network sampling using simulations to generate networks, species ranges,  
19 and biodiversity observations networks (BONs). We followed the approach introduced by  
20 Catchen et al. (2023) to simulate interaction realization and detection, implemented in  
21 `SpeciesInteractionSamplers.jl`.

22 First, we simulate networks with variation across space. To do so, we generate autocorrelated  
23 ranges representing the distribution of species across a simulated landscape and generate  
24 a metaweb using the niche model to define the feasible interactions between the species.  
25 Combining the species ranges and the metaweb information, we obtain the list of **possible**  
26 **interactions** in our landscape and inside every pixel based on the co-occurrence of species  
27 with feasible interactions. Next, we incorporate a consideration for species abundances, which  
28 influence both interaction realization and detection. For example, rare and low-abundance  
29 species are less likely to interact in a location because they are less likely to encounter  
30 one another, especially compared to abundant species, leading to neutrally forbidden links  
31 (Canard et al. 2012; Catchen et al. 2023). We generate an abundance profile for our set of

species (based on a Normalized Log Normal distribution), then draw **realized interactions** at each location given the abundance of species. Finally, we define the set of **detected interactions** by conditioning the detection of realized interactions on the abundance distribution (even if an interaction is realized in a location, we need to be there at that moment to detect, which is again less likely for low probability events; Catchen et al. (2023)).

Second, we design Biodiversity Observation Networks (BONs) across the simulated landscape and measure their efficiency at monitoring the generated networks when increasing the number of sites in the BON. We test different algorithms to spread monitoring sites across the landscape and optimize given different information such as species ranges, species richness or presence of realized interactions. Across our simulations, we used landscapes of 100 x 100 pixels, 75 species, and a connectance of 0.2 to generate metawebs using the niche model.

### *Evaluating metaweb sampling*

First, we investigated the efficiency of sampling networks across space to document a metaweb of interactions, comparing it with the efficiency to document species richness. We generated spatially balanced Biodiversity Observation Networks (BONs) across our landscape, defining sites where to sample species and interactions. We tested all number of sites between 1 and 100 and used the `BalancedAcceptance` method, which aims to distribute sampling evenly across space (for a given number of sampling sites). We then evaluated the sampling efficiency of each BON in terms of overall species richness and compared to the generated metaweb for each interaction type. For species richness, we assumed we sampled all species present in a sampled location, combined this information across all sampled sites to determine the number of species encountered through sampling, and compared the result with the total species richness in the landscape (75 species). For interactions, we aggregated each interaction type separately (possible, realized and detected) for a given BON and evaluated the proportion of the metaweb recovered through sampling (and therefore for a given number of site).

We ran simulations with 100 fully-independent replicates (different metaweb, species ranges, abundance distribution, realized interactions) for each number of site in a BON and for the two interaction references, yielding 10,000 simulations in total. Prior exploratory simulations showed the number of sites in the BON to be the main element affecting the proportion of sampled species or interactions compared to other model parameters such as metaweb connectance and landscape autocorrelation.

### 63 *Focal sampling on single-species & optimization*

64 In complement to the metaweb sampling approach, we also investigated the sampling effi-  
65 ciency on a single-species basis. We simulated a focal sampling where an observer would  
66 target one species and aim to retrieve all of its interactions (here known from the metaweb),  
67 but might have different information to optimize its sampling. Observation network design  
68 and efficiency needs to be evaluated beyond spatial balance, and also regarding their efficiency  
69 at meeting biodiversity monitoring objectives (Norman and Poisot 2025). Thus, the idea of our  
70 focal sampling exploration is to represent a simplified case with a clear objective (retrieving a  
71 species' interactions) and where we do have prior knowledge about the system, either through  
72 empirical data or model predictions, which can be used to optimize sampling.

73 Optimizing focal sampling relies on two elements: the sampler algorithm used and the  
74 information provided for optimization. Species range represents an attainable level of infor-  
75 mation an interested observer could have, for instance based on expert knowledge or species  
76 distribution models. Therefore, we assumed the observer knew the species range across  
77 the landscape (presence or absence). We tested three sampling algorithms for comparison:  
78 simple random sampling (to represent uninformed sampling), weighted balanced acceptance  
79 sampling (aiming to balance sites across space while weighting favorably for species ranges)  
80 and uncertainty sampling (which targets locations with a higher value in the layer provided).  
81 Next, we tested four information layers for optimization: one with the focal species range  
82 (our attainable information knowledge for the observer), one with the probabilistic range  
83 (for a less informed case, similar to the output of a species distribution model), one with the  
84 location of realized interactions for the focal species (to represent a best-case scenario) and  
85 one with species richness across the landscape (to represent a holistic level of information for  
86 the system).

87 Using the approach described in the previous section, we generated a single\* set of metaweb,  
88 species ranges, and locations for realized interactions (\*to showcase a specific example, and  
89 counter-verified with separate sets which yielded conceptually similar results, as described  
90 in the next section). We selected the species with the highest degree (number of interactions)  
91 in the metaweb as the focal species for the simulation. We then generated BONs of 1 to 500  
92 sites (at interval increments of 5 sites) using the three possible algorithms and optimization  
93 layers, with 100 replicates for every number of sites. For each BON generated, we extracted  
94 the number of unique realized interactions sampled across all sites (essentially the monitored  
95 degree for the focal species)

Next, we aimed to verify whether our results were constant when replicating simulations with different simulated metawebs and species ranges. We ran 100 independent simulations, each with 50 replicates for every number of sites (once again from 1 to 500 sites at interval increments of 5 sites). However, doing so raised the issue of how to describe and compare the efficiency of the simulated monitoring across samplers, optimization layers or focal species degree, given how the generated metawebs and landscapes might vary between independent simulations, and especially how to describe the relationship between the number of monitoring sites and the proportion of monitored interactions.

Since the exact proportion of monitored interactions is known in our simulations, we used the following single-parameter equation which generates similar efficiency curves:

$$p = \frac{x}{a + x} \quad (1)$$

where  $p$  is the proportion of monitored interactions,  $x$  is the number of sites in the BON and  $a$  is the efficiency parameter we use for comparison. Here,  $a$  is the single parameter describing the shape of the curve, and can be compared across samplers, optimization layers or focal species degree. The lower this parameter, the faster we reach a high proportion of monitored interactions.

For each independent simulation, sampler and optimization layer option, we performed a grid search across possible values of  $a$  to identify the curve yielding the best fit with the simulation's result. Upon exploratory analysis, we observed that searching for values of  $a$  on an exponential scale allowed for more gradual control over the curve shape, and therefore we searched directly for values of  $e^a$  (effectively replacing  $a$  by  $e^a$  in Equation 1). We performed a grid search of 10,000 evenly spaced values between  $-5$  and  $15$  (limits chosen to go beyond the fitted values), and selected the value of  $a$  yielding the curve minimizing the sum of squared errors to the values from the simulation.

To compare the efficiency curves within a given simulation (i.e., between samplers and optimization layers for a given metaweb and set of species ranges), we used the definite integral of Equation 1 between 0 and  $k$ , the maximum number of sampling sites in the landscape (10,000), which takes the form:

$$\int_0^k \frac{x}{a + x} dx = a \ln(a) - a \ln(k + a) + k \quad (2)$$

Using Equation 2, we can compute the difference in efficiency between two curves as the difference of integrals:

$$d_{1,2} = (a_1 \ln(a_1) - a_1 \ln(k + a_1) + k) - (a_2 \ln(a_2) - a_2 \ln(k + a_2) + k) \quad (3)$$

This difference measure allows to identify which samplers or optimization layer is more efficient, quickly reaching a high proportion of monitored interactions (positive values indicate that the first curve is more efficient than the second), while also quantifying the magnitude of this difference in a comparable way between possible comparison pairs. We compared all samplers and optimization layers individually within each independent simulation, given that the monitoring context is different in each one (different metawebs and species ranges).

## Results

### Lower sampling expectations for species interactions than species richness

Our results for metaweb sampling highlight a marked difference between the proportions of sampled species and interactions (Figure 1). Sampled species saturated very quickly, with all species sampled with less than 25 sites in a BON across the landscape. Possible interactions, which depend on co-occurrence of species with a feasible interaction in the metaweb, also saturated quickly, reaching over 90% of interactions in the metaweb with 25 sites and increasing asymptotically afterwards. In contrast, realized and detected interactions, which account for the impact of species abundances, reach much lower proportions, both under 25% of the metaweb after 100 sites. While the proportion would keep increasing with a higher number of sites, the speed at which interactions accumulate is notably lower than for species richness and possible interactions, highlighting the need for a much higher expected sampling effort.

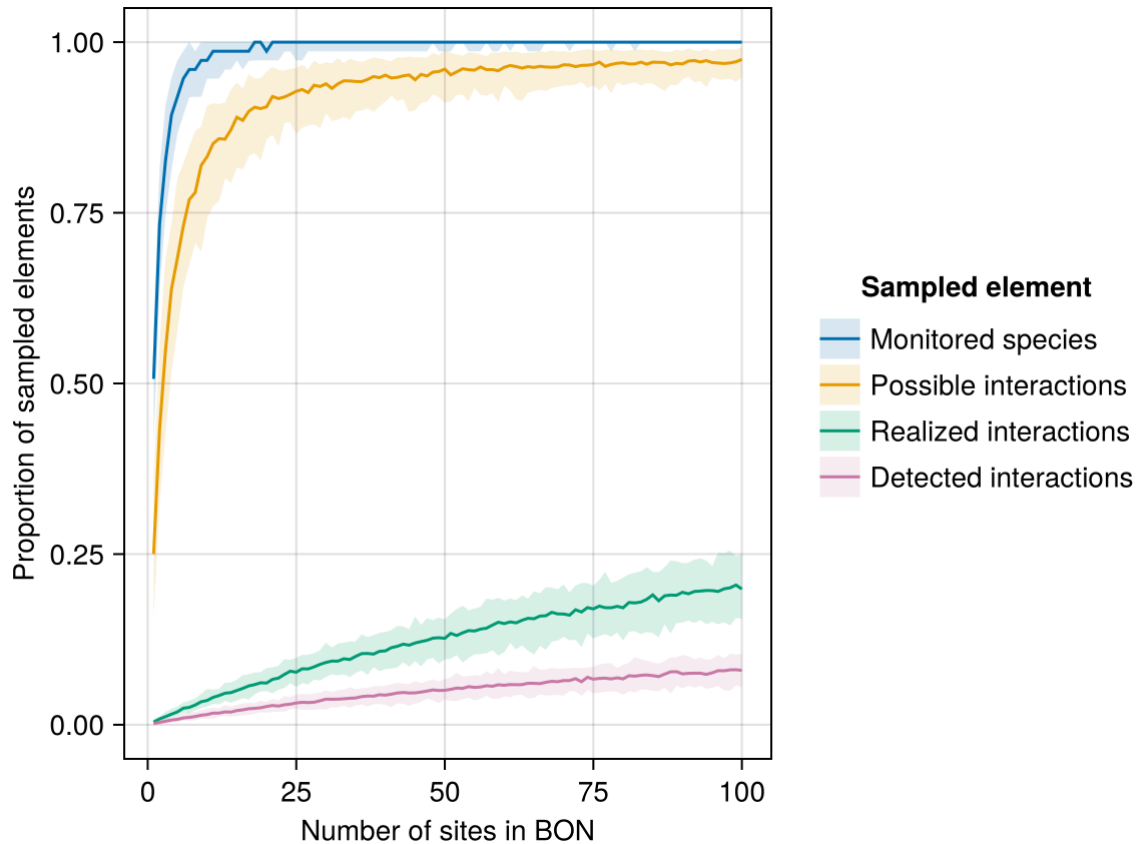


Figure 1: Proportion of sampled species and interactions for a given number of sampling sites in a Biodiversity Observation Network (BON) across the simulated landscape. The proportion refers to the total species richness and the total number of interactions in the metaweb as the maximum reference. The central line indicates the median proportion for a given number of sites while the shaded bands delimit the 5th and 95th percentile across replicates (100 per site, 10,000 in total).

### Differences between samplers and optimization layers

Our focal sampling experiment showed the Uncertainty Sampling algorithm to be most efficient at optimizing BON-design based on a known species range and while aiming to maximize the sampling of interactions of a single focal species (Figure 2A). Simple Random sampling, as a proxy for uninformed sampling, performed the worst among the algorithms tested. Weighted Balanced Acceptance performed notably less than Uncertainty Sampling at maximizing the efficiency for the focal species, although it maintained a better spatial balance across the landscape.

Optimizing on the known species range, our proxy for an attainable level of information an interested user might have, was a more efficient strategy to retrieve the focal species' interactions than optimization based on species richness (or interaction richness), effectively sampling more than half of the species' interactions (Figure 2B). In contrast, optimization

163 based on the location of realized interactions, our best-case scenario requiring a higher level  
164 information, expectedly performed better, retrieving most of the species' interactions.



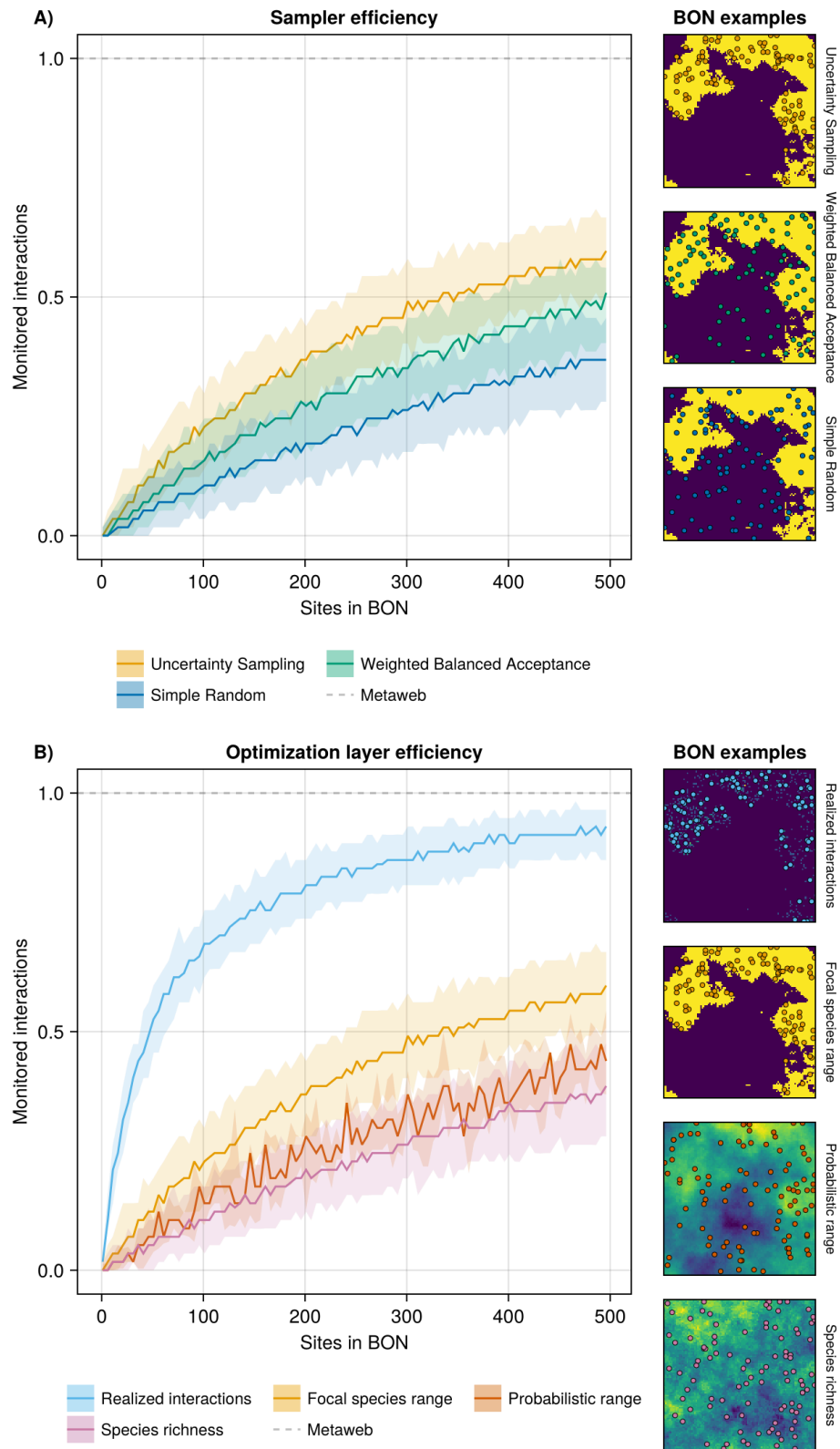
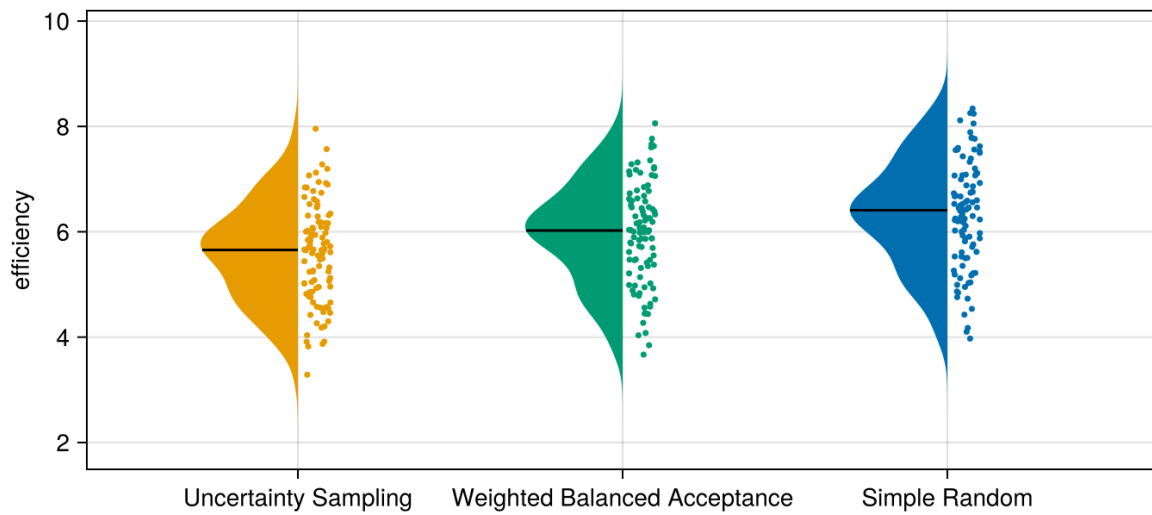


Figure 2: Efficiency of sampler algorithms (A) and optimization layers (B) at sampling realized interactions for a given focal species within Biodiversity Observation Networks (BONs). Panel A compares three samplers used with the species range as the optimization layer while Panel B compares four optimization layers with Uncertainty sampling as the sampler. The central line indicates the median proportion for a given number of sites while the shaded bands delimit the 5th and 95th percentile across replicates. The right panels illustrate examples of BONs of 50 sites (coloured points) generated by each sampler over the landscape (species present at pixels in yellow).

### **Comparison of efficiency parameters across simulations**

The distribution of efficiency parameters across independent simulations confirmed the interpretations based on Figure 2, but also highlighted a wide range of outcomes in efficiency between simulations. Among samplers, the median efficiency parameter indicated a better performance (lower parameter value) for Uncertainty Sampling, followed by Weighted Balanced Acceptance and Simple Random (Figure 3A). The actual efficiency values spanned a similar range across the possible sampler options with all simulations combined. Among optimization layers, optimizing on the focal species range performed better than on the probabilistic range, followed by species richness, although all three options spanned similar ranges with all simulations combined (Figure 3B). In contrast, optimizing based on the realized interactions distinctly and consistently performed better across all simulations.

### A) Samplers



### B) Optimization Layers

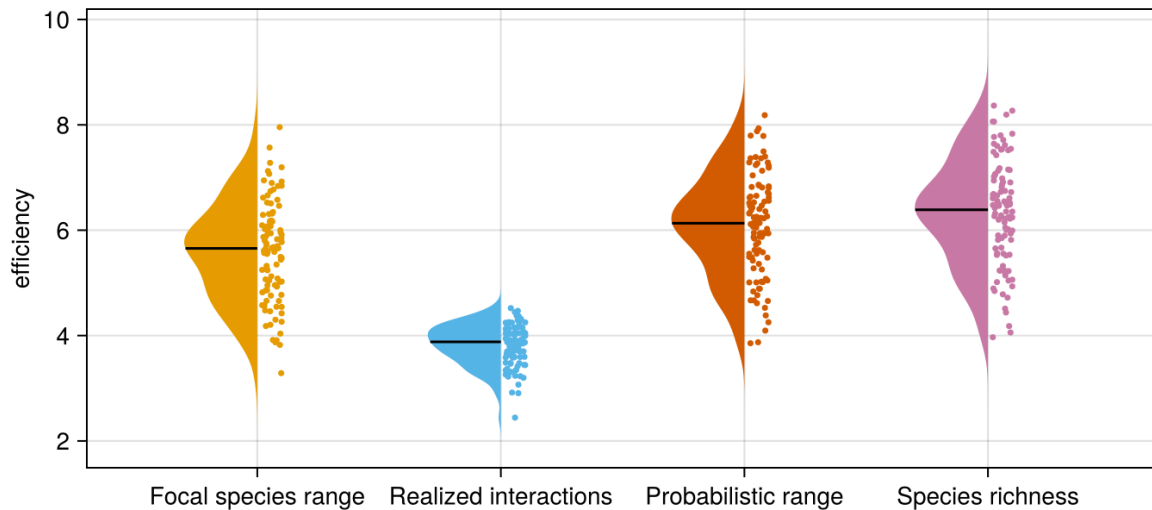


Figure 3: Distribution and density of efficiency parameters across 100 independent simulations for the three sampler options tested on the focal species range (A) and for the four optimization layer options using Uncertainty Sampling as the sampler (B). Uncertainty Sampling on panel A and Focal species range on panel B therefore represent the same data.

The fact that multiple options span similar ranges of efficiency values across independent simulations highlights the need to compare the efficiency within a given simulation. Comparing options one-by-one clearly highlighted a most efficient one in all case (Figure 4, Supp. Mat.), consistent with the showcased single-simulation example (Figure 2). Uncertainty Sampling was more efficient than Weighted Balanced Acceptance in 98 of 100 simulations, and both were more efficient than Simple Random in all simulations (Figure 4). Optimizing on the focal species range was always more efficient than on the probabilistic range or species richness

196 (with probabilistic range more efficient than species richness in all but one simulation), but  
197 always less efficient optimizing on realized interactions.

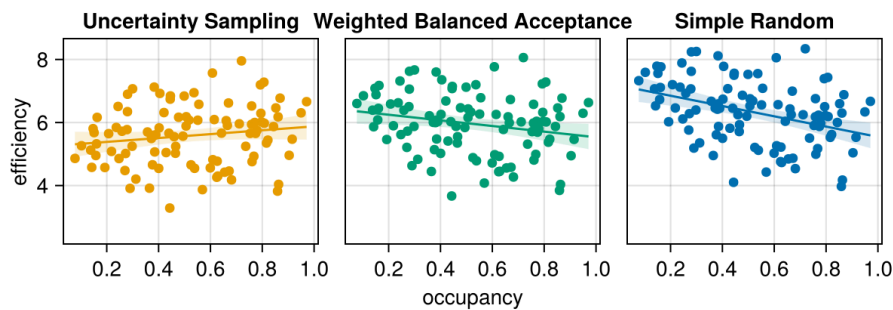


198  
199 Figure 4: Within-simulation comparison of efficiency between samplers and optimization layers. The  
200 comparison values is based on the difference of the definite integrals of the efficiency samplers curves  
201 described in Equation 3. Negative values (in blue) indicate that the compared option lead to a less efficient  
202 sampling than its reference, while positive values (in orange) indicate it was more efficient. References were  
203 chosen for narrative purposes and to simplify the number of comparisons displayed. All one-by-one  
204 comparisons are available in Supp. Mat.

205 While the simulations clearly highlight which sampler or optimization layer is most efficient  
206 across all simulations, efficiency parameters and differences vary over a wide range of  
207 values. We verified if this could be tied to the occupancy of the focal species in simulated  
208 landscape, measured as the number of cells where the species is present. We examined the  
209 relationship between occupancy and efficiency. Optimizing using Uncertainty Sampling on  
210 the focal species range and on realized interactions is especially efficient at low occupancy, as  
211 it targets the fewer cells where the focal species is present. As species occupancy increases,  
212 the difference between options decreases. When the focal species has high occupancy and  
213 is present almost everywhere across the landscape, Weighted Balanced Acceptance and  
214 Simple Random perform similarly to Uncertainty Sampling, as balancing sites across space

or choosing at random becomes equivalent to targeting the species range. Optimizing based on species richness became more efficient as occupancy increased. A possible explanation is that at high occupancy, species range is not a restricting element and thus targeting sites with high richness directly leads to targeting sites with more interaction partners. On the contrary, at low occupancy, species range is an important restriction, and optimizing on richness of all species might target sites where the focal species is absent. Optimizing on the probabilistic species range behave similarly to species richness; at high occupancy, it is very likely the species will be present at the sites with high probabilities and therefore the sampler can more effectively filter out sites where the species is absent. At low occupancy, the focal species will be present in fewer locations with high probabilities, but might be absent from locations with intermediate probabilities, making it more challenging for the sampler to pick apart.

#### A) Samplers



#### B) Optimization Layers

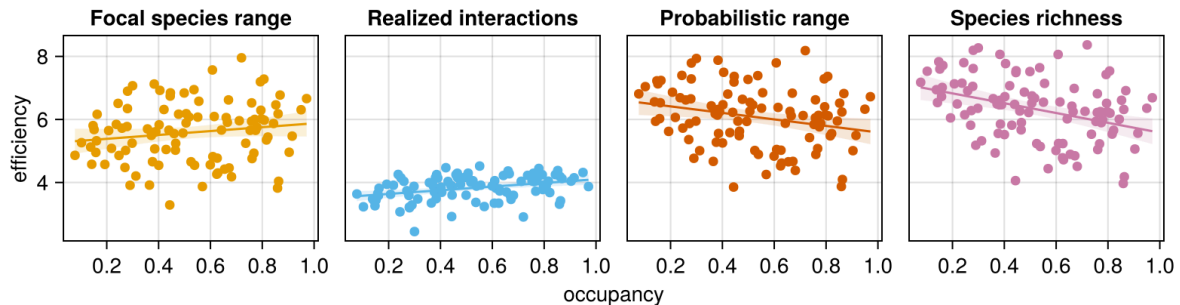


Figure 5: Relationship between the occupancy of the focal species and sampling efficiency for all optimization options. These results come from 100 independent simulations, and therefore 100 different sets of metawebs, species ranges, and occupancy of the focal species.

## Discussion

What should be realistic expectations for sampling and monitoring of species interactions? Our simulations highlight they should be much lower than for species richness once considering that interaction realization and detection depend on species abundance. This means we

can expect the monitoring of species interaction within a BON to require a higher monitoring effort than the one required for monitoring species and community composition.

We can improve coverage with the right optimization method, but efficiently doing so requires a clear and attainable sampling or monitoring objective. For instance, adopting a single-species perspective can yield relevant results while relying on attainable levels of information. Optimizing on a known species range proved an efficient strategy to retrieve its interactions, better than optimization based on more holistic-information such as species or interaction richness. Optimizing based on the location of the realized interactions was even more efficient, but our simulations represent a best case scenario likely beyond reach in an applied context. Nonetheless, it highlights that the most important efficiency gains will come from better knowledge of where interactions are realized. Better incorporating considerations for species abundance, the factor influencing interaction realization in our simulations, will be crucial and potentially more impactful than improving knowledge or reducing uncertainty over species ranges.

When does the choice of sampler or optimization layer matter the most? Here we concur with Norman and Poisot (2025) that ecologically-relevant metrics are required to evaluate the performance of monitoring designs, beyond spatial balance. However, while they found that many sampling algorithms performed similarly for biodiversity monitoring, our results highlighted better options when the monitoring target is highly localized. We observed the most important differences for species with low occupancy, where Uncertainty Sampling is relevant to target only the focal species range. Differences were less important at higher occupancy and for species present across the whole landscape, although Uncertainty Sampling still performed better. This result should be expected, as a specialized sampler should be more efficient when a species range is highly localized compared to one balancing sites across the the whole landscape. However, it also applies in broader context, highlighting how some monitoring targets (e.g., species interaction) expected to be highly localized or more likely in specific locations might benefit from targeted monitoring designs. In such case, efficiency gains will depend on improvements in our capacity to predict the location of those phenomena and on our ability to continuously evaluate and update predictions based on the information acquired.

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